EX No : 3	DATA CLEANING AND DATA TRANSFORMATION

Aim

To analyze the Sales Data Analysis dataset, perform data cleaning and transformation, calculate derived sales value, visualize product and city performance, build a Random Forest Regression model to predict sales, evaluate the model, and export the processed data to Excel.

Procedure

1. Load the Sales Data Analysis dataset using Pandas.
2. Display the first few records, shape, information, and missing values.
3. Convert the Date column into datetime format.
4. Check and handle missing values.
5. Create a Sales column using Price × Quantity.
6. Perform EDA using descriptive statistics and sales distributions.
7. Analyze sales by Product and City.
8. Prepare numerical and categorical features for machine learning.
9. Split the dataset into training and testing data.
10. Train a Random Forest Regression model to predict Sales.
11. Predict sales for the test dataset.
12. Evaluate the model using MAE, RMSE, and R^2 score.
13. Export the processed dataset to Excel.

Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
```

```
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# 1. Load dataset
df = pd.read_csv("9. Sales-Data-Analysis(1).csv")

# 2. Display data and information
print(df.head())
print(df.shape)
print(df.info())
print(df.isnull().sum())

# 3. Convert Date
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

# 4. Create Sales target
df["Sales"] = df["Price"] * df["Quantity"]

# 5. EDA
print(df.describe())
print(df.groupby("Product")["Sales"].sum().sort_values(ascending=False))
print(df.groupby("City")["Sales"].sum().sort_values(ascending=False))

# 6. Visualization
df.groupby("Product")["Sales"].sum().sort_values(ascending=False).plot(kind="ba
r")
plt.title("Total Sales by Product")
plt.xlabel("Product")
plt.ylabel("Sales")
plt.xticks(rotation=35)
plt.show()
```

```
df.groupby("City")["Sales"].sum().sort_values(ascending=False).plot(kind="bar")
plt.title("Total Sales by City")
plt.xlabel("City")
plt.ylabel("Sales")
plt.xticks(rotation=35)
plt.show()

# 7. Date features
df["Year"] = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
df["Day"] = df["Date"].dt.day

# 8. Features and target
X = df.drop(columns=["Sales", "Date", "Order ID"])
y = df["Sales"]

num_cols = X.select_dtypes(include=np.number).columns.tolist()
cat_cols = X.select_dtypes(exclude=np.number).columns.tolist()

# 9. Preprocessing
preprocessor = ColumnTransformer([
    ("num", SimpleImputer(strategy="median"), num_cols),
    ("cat", Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("onehot", OneHotEncoder(handle_unknown="ignore"))
    ]), cat_cols)
])

# 10. Random Forest Regression
model = Pipeline([
    ("preprocessor", preprocessor),
    ("regressor", RandomForestRegressor(
        n_estimators=200, random_state=42
    ))
])
```

```
# 11. Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
# 12. Train
model.fit(X_train, y_train)

# 13. Predict
y_pred = model.predict(X_test)

# 14. Evaluation
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)

# 15. Export
df.to_excel("Sales_Data_Analysis_Processed.xlsx", index=False)
print("Analysis completed!")
```

Output

First five records:

Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
10452	2022-11-07 00:00:00	Fries	3.49	573.07	Online	Gift Card	Tom Jackson	London
10453	2022-11-07 00:00:00	Beverages	2.95	745.76	Online	Gift Card	Pablo Perez	Madrid
10454	2022-11-07 00:00:00	Sides & Other	4.99	200.4	In-store	Gift Card	Joao Silva	Lisbon

10455	2022-11-08 00:00:00	Burgers	12.99	569.67	In-store	Credit Card	Walter Muller	Berlin
10456	2022-11-08 00:00:00	Chicken Sandwiches	9.95	201.01	In-store	Credit Card	Walter Muller	Berlin

Dataset Shape: (254, 10)

Missing Values: No missing values were found in the supplied dataset.

Total Sales: 769,515.86

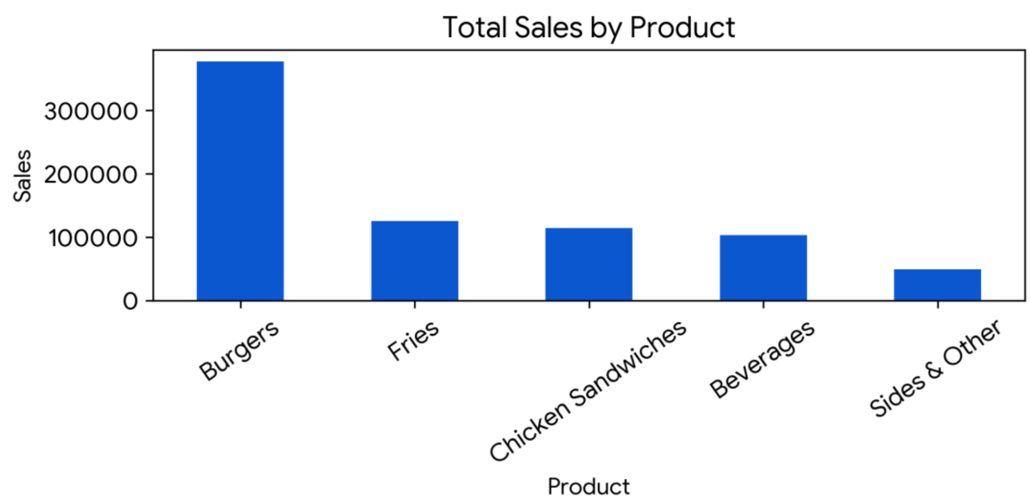
Minimum Sales: 1,000.00

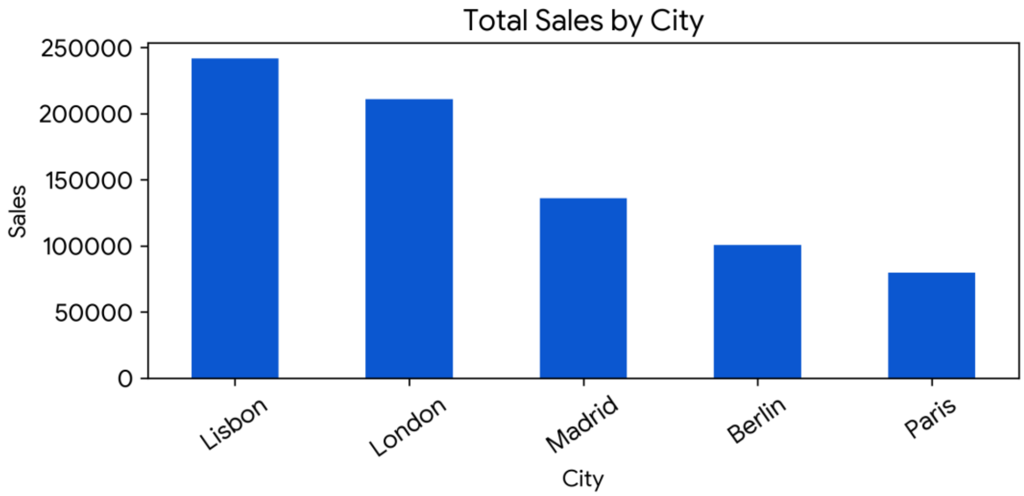
Maximum Sales: 16,074.43

EDA Results

Highest total sales by product: Burgers (376,999.81)

Highest total sales by city: Lisbon (241,714.12)





Model Evaluation

Training records: 203

Testing records: 51

MAE: 37.4321

RMSE: 95.7656

R^2 Score: 0.9983

Result

The Sales Data Analysis dataset was successfully imported, explored, processed, and modeled. The dataset contains 254 records and 9 original fields. A Sales target was derived as Price × Quantity. EDA identified Burgers as the highest-sales product and Lisbon as the highest-sales city. A Random Forest Regression model was trained and achieved an R^2 score of 0.9983 on the test data. The processed dataset was exported successfully to Sales_Data_Analysis_Processed.xlsx.